CHARACTER FILM / 14 S LOOP / 16:9 / 30 FPS

One ramp.
Zero brakes.

The mascot floors a red roadster through a neon tunnel, bursts into a canyon of towers, flips its pixel shades down and launches off a ramp over a missing bridge span. Time slows to six percent in the air, the landing slams on the beat in a fountain of sparks, and the next tunnel is the first frame again. Every frame is a Blender scene written by Claude Opus 5.5 from the prompt on page 2.



14.0 s	30 fps	1920x1080	3 files	320	4 + 2
SEAMLESS LOOP	420 FRAMES	CYCLES ON THE GPU	SCENE, TIMELINE, SOUND	WORDS OF PROMPT	FOLLOW-UPS, PICTURE + SOUND



0.63 S
THE TUNNEL



2.47 S
BREAKOUT



4.97 S
THE RAMP



6.47 S
SLOW-MO



8.83 S
THE SLAM



13.30 S
LOOP HOME

THE FORMULA

01 One world, repeated
Tunnel, highway, gap and towers are one segment instanced four times, so the loop needs no cut.

02 Beats set the physics
The tempo is chosen so 8 bars fill 14 s; the lip and the landing sit exactly on beats 12 and 20.

03 Slow-mo in space, not time
The air time remaps distance, never gravity, so the arc keeps its shape while the clock slows to 6%.

04 The mascot from its logo
Every voxel of HERO comes from the terminal logo grid: 12x8 body, slit eyes, stub arms, four legs.

05 Fill the face, not the city
A small light rides in the car aimed at HERO; the neon only has to paint reflections.

06 Measure, don't guess
Seam checked as a frame difference, sound as a spectrogram, loudness with ffmpeg ebur128.

PAGE 02
The prompt, copy-ready, block by block.

PAGE 03
Every follow-up, word for word, in order.

PAGE 04
Make it yours, render it fast, every beat on its frame.

PAGE 05
Sound, its prompt and both follow-ups.

The prompt

COPY-READY 320 WORDS

Paste it as is into Claude Code running Opus 5.5 at effort high, with Blender 5 and ffmpeg installed. It writes the scene as Python, renders it headless on the GPU and checks its own stills.

```
BRAND
HERO: voxel mascot 18x10, body 12x8 at (3,0), arms 2x2 at (1,4) (15,4), legs 1x2 at x 4 6 11 13 row 8, eye slits 0F0F0F 1x2 at (5,2) (12,2), 6 deep, beveled, orange D97757
RIDE: open roadster, low wedge nose, full-width tail light, candy red 6B0A0F clearcoat, three hidden rear thrusters, orange underglow
PROP: pixel "deal with it" shades, white glints, parked on HERO's head
SET: neon tunnel through a megastructure, then a highway in a night canyon of towers, kicker ramp, missing bridge span, wet asphalt
SIGNS: "CLAUDE CODE" on the tunnel wall, giant pixel-HERO billboard, ROAD CLOSED gantry
PALETTE: cyan 14B4FF, magenta FF1A78, ring FF6124, sodium FFB873, sky 0D0612

FORMAT: Blender 5 Python, Cycles GPU, 1920x1080, 30 fps, 14.0 s seamless loop: the world repeats every period P so frame 420 hands off to frame 1. One timeline.py drives picture and sound; 137.14 BPM, 8 bars = 14 s.

BEATS (s):
0 HERO drives at the lens through the tunnel, shades up, rings rushing past
2.1 bursts out into the canyon
4.75 hits the ramp; 4.95 shades flip down
5.25 launch on the beat: thrusters fire, speed ramps to 6% slow-mo
6.0 arc over the gap, arms up, slight roll, billboard behind
8.75 slam on the beat: suspension dips, sparks, camera shake
12.1 shades flip back up; 12.57 into the next tunnel = frame 1

CAMERA: one take locked to the car: front low 3/4, swing wide at the exit, chase on the ramp, orbit over the canyon, front low for the slam, settle on the opening offset.

LOOK: puddles mirror the neon, mist fog, bloom, slight dispersion, shallow DOF on HERO, 0.5 shutter blur; a soft fill rides with the car so the face reads D97757.

CHECKS: preview 9 beat stills at half res before the full render; eyes readable in every front shot; frame 420 to 1 no bigger a jump than any neighbour.
```

BRAND	**The block you swap.** HERO, RIDE, PROP, SET, SIGNS, PALETTE. The rest of the prompt only uses these names.
FORMAT	A Blender script, not a hand-made file: a pure function of the timeline, rebuilt from scratch on every change.
BEATS (s)	The ride in seven lines: tunnel, breakout, ramp, launch, flight, slam, back into the tunnel.
CAMERA	One take that rides with the car and ends exactly where it started, so the loop has no seam.
LOOK	Wet neon city at night, and the one color that must survive red paint and pink light.
CHECKS	Makes Opus look before it spends an hour on the GPU. Most of the polish comes from this line.

Every follow-up

4 ROUNDS

After each look at the stills, a render log or the final file I wrote what I saw and what I wanted next. Word for word, in order. The first build came from the prompt alone and already had the tunnel, the jump and the loop.

FOLLOW-UP 1 / AFTER THE FIRST 9 PREVIEW STILLS

```
HERO reads as a pile of bricks on the back seat: at 0.08 per voxel it is a third
of the car. Make the voxel 0.105, sit HERO in the middle of one bench seat and
bring the opening camera to 5.4 m, aimed at its eyes, not the bonnet. The neon
pixel-HERO billboard blows out the whole top of every jump frame: drop it to a
third of the brightness, shrink it and set it 6 m back on its tower. The windows
are 5 m tall, so the towers read as toys next to the car: make them about 1.6 by
2.7 m. At the slam the sparks are invisible: 240 of them, thicker and twice as
bright.
```

FOLLOW-UP 2 / AFTER A FULL-RES CROP OF HERO

```
The steering wheel and the blue dash screen read as a white disc floating over
HERO's face. Take both out, a speedster does not need them on camera. HERO sits
so low the cowl hides its eyes: lift it 10 cm on a thicker seat and drop the
windscreen to a low wraparound so both eye slits clear it in the front shots.
```

FOLLOW-UP 3 / AFTER TIMING THE FIRST FULL-QUALITY FRAMES

```
Each 1080p frame takes 2.5 to 3.5 minutes: that is the CPU, because the GPU
choice lives in the user preferences, not in the .blend. Write a gpu.py that
switches Cycles to OptiX and pass it on every render. Then time one frame again
before you start all 420.
```

FOLLOW-UP 4 / AFTER THE FINAL ENCODE

```
The loop is clean, but the MP4 is 118 MB at 67 Mb/s: the per-frame grain is noise
the encoder cannot share between frames. Drop the grain, keep the vignette, CRF
18.
```

FOLLOW-UP 1

Result: HERO fills a third of the frame in every front shot; the city reads as towers.

FOLLOW-UP 3

Result: OptiX: 7.3 s a frame, 420 frames in 54 min instead of about 20 hours.

FOLLOW-UP 4

Result: 29 MB; the seam jump (23.4) sits inside the neighbour range (20.5 to 26.1).

FOLLOW-UPS 1 AND 2 IN ONE PICTURE: THE OPENING FRAME



Before (first build, 0.0 s): a small brick stack behind the windscreen, the face hidden by the cowl.



After (0.0 s): HERO fills the cockpit, slit eyes clear the glass, shades parked on its head.

Make it yours

Only the BRAND block changes. Your hero, your ride, your city. It works for any "we shipped it" or launch-day post.

```
BRAND
HERO: your mascot as a pixel grid from its logo, its hex colors
RIDE: voxel hover-bike, white F2F0EA panels, cyan 2EC5D3 vents, two jet pods
PROP: a scarf that flaps behind it, mint 7FD8B6
SET: snowy mountain pass at dawn, an ice tunnel, a broken rope bridge over a gorge
SIGNS: your logo carved in a glacier wall, trail markers
PALETTE: dawn FFC9A8, ice 9FD7F0, rock 3C3A45, snow F6F7FA, sun FF8E5C
```

Render it fast

1. Build the .blend from scene.py with `blender -b --factory-startup -P scene.py`.
2. Render 9 beat stills at 50% and 48 samples first; read them as a contact sheet before anything else.
3. Pass `-P gpu.py` on every render: the OptiX device is a user preference, not part of the file.
4. Full render at 1920x1080, 128 samples adaptive, OptiX denoise, 0.5 shutter, persistent data: about 7 s a frame on an RTX 4070 SUPER.
5. Encode H.264 High CRF 18 at 30 fps with a soft vignette, mux the WAV as AAC 320k, +faststart.

Every beat on its frame

Checked on the rendered frames at 30 fps. The tempo is 137.14 BPM, so every hit below also lands on a 16th note of the music.

TIME	FRAME	PICTURE	SOUND
0.00 to 2.10 s	f1 to f64	HERO drives at the lens, rings rush past four times a second	engine roar, ring whups, tunnel reverb
2.14 s	f65	bursts out of the tunnel into the canyon	big whoosh, the reverb opens up
3.00, 4.35 s	f91, f131	camera swings wide, then chases from behind	two pass-by whooshes, snare roll, riser
4.75 s	f143	front wheels hit the ramp	steel clunk
4.95 to 5.25 s	f149 to f158	shades flip down onto the eyes	a bright shing
5.25 s	f158	launch: thrusters fire, time ramps down	BRAAAM, jet roar, music drops into a low pass
5.70 to 7.40 s	f172 to f223	apex at 6% speed, arms up, the billboard behind	squeaky wheee, engine dives, heartbeat kicks
8.75 s	f263	slam: suspension dips, sparks, camera shake	sub boom, metal crunch, sparks, tyre screech, crash
12.12 s	f364	shades flip back up	small click
12.57 to 14.0 s	f378 to f420	into the next tunnel, the opening frame again	whoosh, reverb closes, bar 1 comes round

3 TIPS

1

Repeat the world

Build one road segment and instance it. Frame 1 and frame 421 are then the same place, so the loop is free.

2

Put the jump on the beat

Pick the tempo from the loop length, then place the lip and the landing on beats.

3

Slow distance, not gravity

Remap how far the car has flown, keep the arc. Slow-mo then never bends the jump.

Sound

COPY-READY 210 WORDS

The picture is a Blender scene; the sound came from one more prompt. Every sound is synthesized in numpy, driven by the same timeline.py as the picture.

```
Add sound. Synthesize everything in Python with numpy, scipy and pedalboard: no samples, no voices, no words. 48 kHz stereo WAV, 14.0 s, read every event from timeline.py and land each hit on its frame. Render three loops and keep the middle one so every tail wraps into the start.

Music: a dark action trap cue at 137.14 BPM in D minor, 8 bars that loop. 808 on the root with glides, hard kick, clap snare on 3, hats with rolls into bar ends, a 16th-note pluck ostinato in a 3-3-2 accent. Bar 3 builds: snare roll, filter sweep, riser into the lip. On the launch a low BRAAAM, then slow-mo: everything low-passed, a pad, heartbeat kicks, a reverse cymbal into the landing. The beat drops back on the slam.

Car: an engine whose pitch follows the speed and dives in the slow-mo, a thin electric whine, a whup for each tunnel ring and tunnel reverb only while inside. Whooshes on the camera swings, a clunk on the ramp, a shing for the shades, a jet roar for the thrusters, a squeaky wheee from HERO, then the slam: sub boom, metal crunch, spark crackle, a short tyre screech.

Master loud, like a trailer: glue compressor and a limiter at -1 dBFS.
```

Every sound follow-up I sent

SOUND FOLLOW-UP 1 / AFTER READING THE SPECTROGRAM

```
Everything from 20 to 200 Hz is one solid wall for all 14 s and the loudness is flat at -9 dBFS RMS from start to end: the engine and the 808 bury the kick and there is no dip in the slow-mo. Halve the engine, high-pass it at 70 Hz, drop it 80% while airborne, duck the bed under the launch and the slam, and keep the slam and the BRAAAM on their own bus. The seam also jumps 0.2 between the last and first sample: the engine's half-speed firing term and its 3.02x partial do not complete whole cycles in 14 s. Round the engine to an even cycle count and use 3.0x.
```

Result: the seam jump falls to 0.05, inside the normal sample-to-sample step of 0.056.

SOUND FOLLOW-UP 2 / AFTER THE SECOND SPECTROGRAM

```
Still too much below 60 Hz under the slow-mo: take the D1 out of the BRAAAM and start it at D2, high-pass the 808 at 38 Hz and bring it down a third, and high-pass the master at 32 Hz.
```

Result: -9.2 LUFS integrated, loudness range 2.6 LU, peak -0.6 dBFS, no clipped samples. Trailer-loud on purpose; for a platform at -14 LUFS just lower the master 5 dB.

Into the MP4: the WAV was muxed in as the video's AAC 320k track with ffmpeg; the same step works for your own video.